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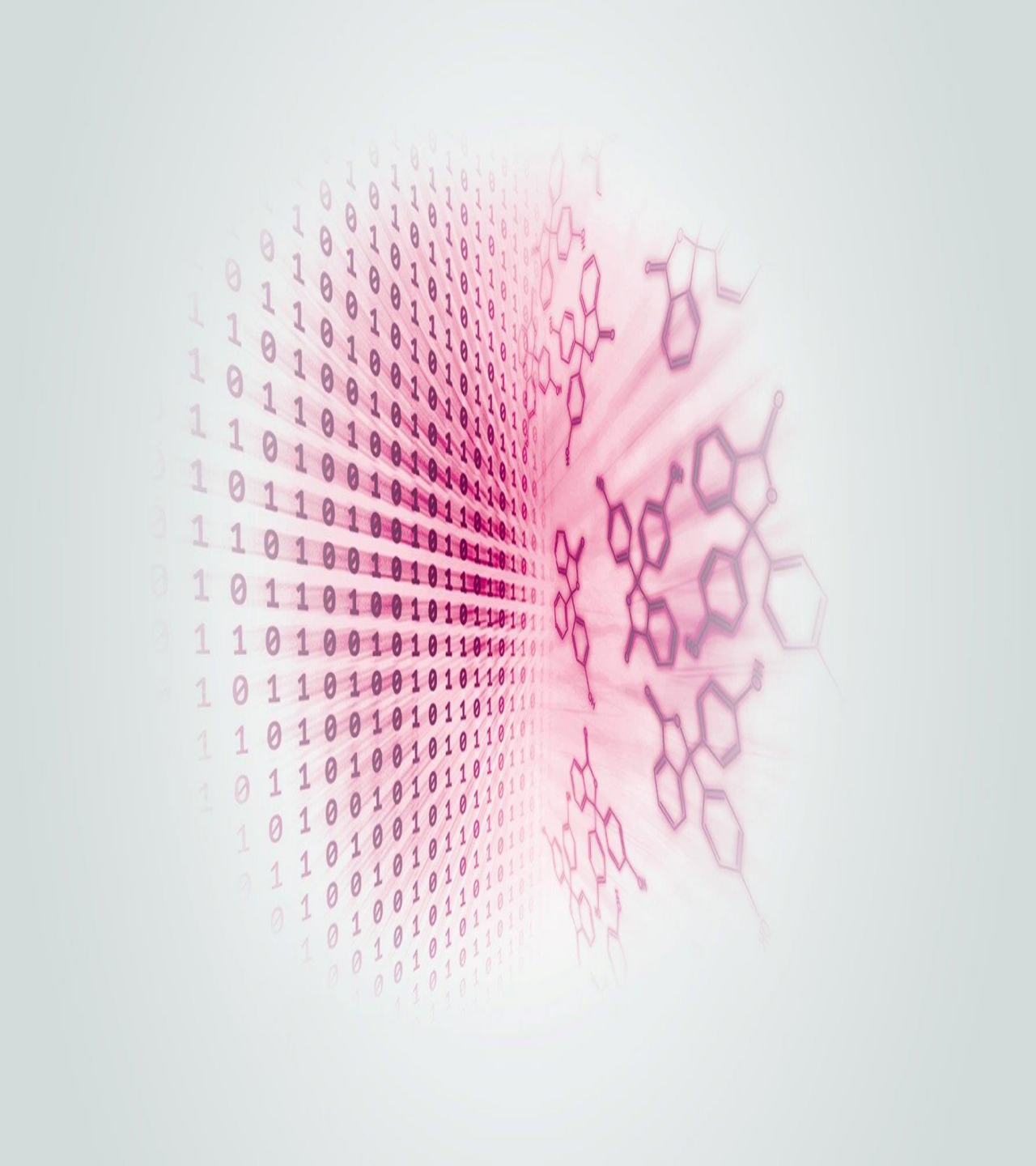
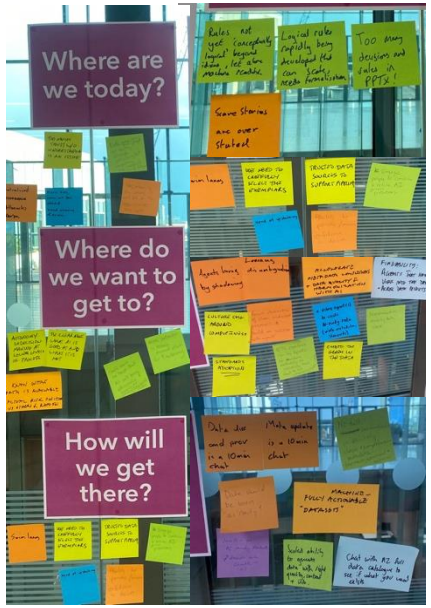
ACCELERATING THE USE OF AGENTIC AI

Envisioning the future of Agentic in R&D Data and AI Governance at AstraZeneca

Workshop Writeup





Thanks for helping shape the R&D Data Office's Agentic AI Journey at AstraZeneca

We appreciate your time, insights, and dedication during Friday's workshop.

Together, we are at the start of a transformative journey, one that will shape our path to achieving the North Star vision for our data office.

Your contributions today have set the foundation for the changes we need to implement. As we move forward, remember that this is only the beginning – your leadership and commitment are key to driving this success.

In the following document, you'll find the ideas we collectively developed that will guide you in your next steps.

Thank you for your partnership and leadership.

Thanks to everyone who has shaped the journey and will be core to making an impact in the months ahead

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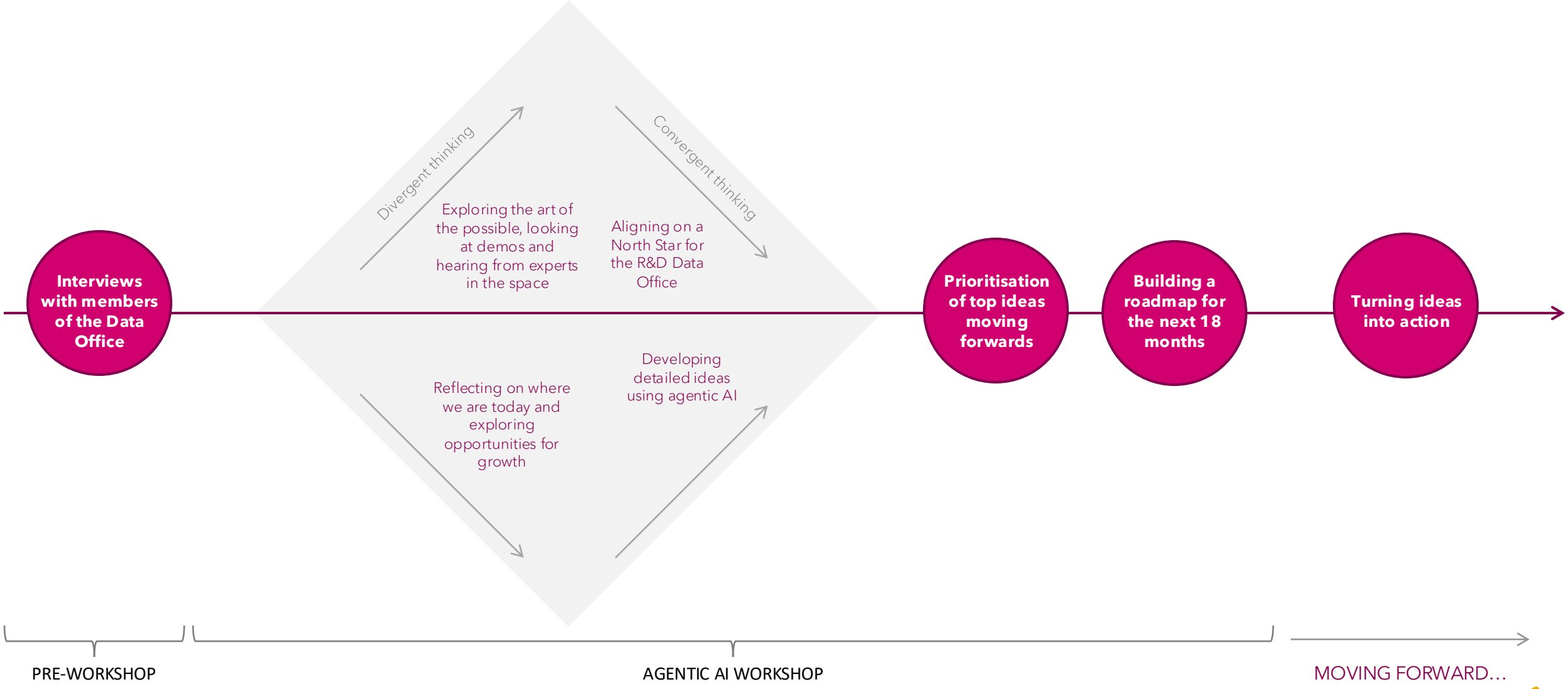
OUR APPROACH



What's in this pack?

1. **Mirror**
   An overview of the key themes arising from the interviews, with reflections on where we are today, where we want to go, and the journey to get there

2. **Art of the Possible**
   Slides shared by SMEs around the use of agentic AI, as well as a summary of the demos shared

3. **North Star**
   The shared North Star vision created by you, using AI to collate reflections and iterated based on feedback

4. **Ideation**
   The ideas created by all three groups, around Data & AI Governance and Policy, Access – 90/10, and Standards & Interoperability

5. **Prioritisation and Roadmap**
   The top voted ideas placed on a visual depicting the roadmap over the next 18 months, with owners for each idea identified

Mirror

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Data Access & Approvals Latency

We're only about 20% along the journey to making access seamless. The big blockers are finding the data, finding the owner, and getting someone willing to take accountability.

I want to live in a world where data flows through without me ever touching it - access, compliance, all handled automatically.

We've been pushing towards a paradigm shift where 90% of data is accessible by default, and only 10% requires special approval.

Imagine if access was just provisioned automatically based on role - that's where we want to get.

- Data access slowed by lengthy governance and review cycles.
- Processes for approvals are tedious, inconsistent, and delay science.
- Large volumes of data exist, but unclear which study contains what.
- Stakeholders want 90–10 access model (90% by default, 10% exception).
- Need better study/data catalogues for faster discovery.

Culturally, AstraZeneca struggles with data access. There are a lot of feelings and few facts guiding how decisions are made.

Right now, if you want access, it's a long process of requests and reviews. By the time you get approval, the opportunity has often passed.

Sometimes it feels like we're all chasing ghosts - we know the data exists somewhere, but no one can point us to it.

It would be a real unlock if we stopped treating data as 'mine' and started treating it as an enterprise asset.

Potential Use Cases:

- Policy-aware, role-based access provisioning.
- Pattern-based access approvals (learning from similar profiles).
- Temp scoped tokens for time-bounded access.
- Agent identity & entitlements: define what an agent can do.
- Automated audit trails of data access and usage.

Where are we today?

99% of time waiting on data wrangling	Variable responses from data owners to risk	Processes like iDap have many steps and would be difficult to automate even with agents	A lot of info and many steps to make a decision	Access requests are use case by use case; access provisioning is slow and without the utilization of patterns and building on previous decisions; process repeated again and again with a lot of manual work	Access to RWD expires, unpredictable how long you will have access to it which limits use for AI	Some stakeholders don't see the value in sharing their data with others - what's in it for them? Eg. preclinical imaging.
We struggle with access but how bad are we competitively?	Framework exit turn this into full practices	Why collect the data if we cant use it? Or allow access?	Inconsistent definition of data governance; opportunity for consistency and automation	Internal / External data sits in different places, with difference governance models.	Data access slowed by governance & reviews; approvals are inconsistent, delaying science and wasting opportunities.	

Where do we want to get to?

Be able to ask for data of a particular type (endpoint, lab, modality), get recommended what studies include that data, and provisioned in one interface.	a consistent workflow that builds on previous experience without a lt of manual intervention and decisions that are subjective and subject to change	90–10 access model: 90% of data accessible by default, 10% requiring special approval, with seamless role-based provisioning.	One stop shop?	Access should be mins away (or a reason if not)	A shift in risk / reward balance – only truly sensitive restricted
Data available off the shelf without request and no/short turnaround	Data openly available for use for right users @ right time	A simpler, scalable reusable approach to open access for data	No critical path access approvals in 90%	Access by default with easy to understand guardrails enabled by agents	

How will we get there?

utilizing AI/tech to guide the decision process, learn from previous decisions, enable human validation for edge cases; keep track with external regulations,	Implement policy-aware, role-based access with automated approvals, temp tokens for time-bound access, and audit trails.	Non-inferiority eval of high risk 'good enough' threshold	Automate DeID	Valuable use cases as exemplars	For 10% truly sensitive role based access
Accelerate productive AI-readiness Data Build differentiation assets	Development of new access policies Leverage enterprise tech and strategic platforms/framework	Democratise access from top – caution where truly merited	Iterations of data and user scope to 90/10, alongside tools, agents, scale to clients	Hybrid	

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Metadata, Standards & Data Readiness

We are starting to see teams pilot metadata automation - and the results are promising. It's showing people what's possible.

We're still in a world where completeness of information is optional. In the age of AI, that can't be negotiable.

Imagine if every dataset came with a complete blueprint - metadata, ontology links, consent flags, everything. That's the goal.

There's optimism that we can close the readiness gap quickly - if we agree on standards and enforce them consistently.

- Metadata and standards inconsistent; FAIR principles not applied uniformly.
- Repeated curation work due to fragmented vocabularies.
- Consent gaps and sovereignty rules limit reuse.
- Desire for AI-ready datasets with enforced standards.
- Synthetic data seen as solution where gaps exist or consent is missing.

Our data scientists spend 60% of their time wrangling data into usable form - AI-ready pipelines would flip that ratio.

Synthetic data could be a game changer - filling gaps where consent isn't there or where the real data doesn't exist.

Knowledge graphs and semantic fabrics are exciting. They give us a way to connect context across studies.

Right now, FAIR is talked about everywhere but not applied uniformly in practice.

Potential Use Cases:

- Automated metadata & lineage tagging (ontology links, synonyms refreshed).
- Pre-approved blueprints for analysis-ready provisioning.
- Standards enforcement across functions.
- Knowledge graph / semantic fabric to encode domain context & enable reuse.
- Synthetic data generation from past study patterns.

Where are we today?

Need to contextualize Fair Data Principles across the data offices (not speaking specifically about R&D)	Is this true? Doesn't it depend on the AI?	Metadata & standards inconsistent; FAIR principles applied unevenly; repeated curation slows reuse and creates fragmentation.	'F' of FAIR already a blocker
Limited interest/ commitment from data owners to making data FAIR	Recognition of what is required through the 'gap' between now and then takes commitment	Metadata is incomplete and inaccurate	

Where do we want to get to?

Harmonization and standardization of data across multiple datasets and origins of data.	FAIR and AI ready by design data generation (internal and external data)	Baseline requirement on how does good data look like which can be utilized by pipelines/agents to model, harmonize curate data; building on the same semantic infrastructure	Every dataset AI-ready: standardized metadata, consistent ontologies, and synthetic data generated ethically to fill gaps.	Scaled ability to generate data with right quality, context and info	Fully machine-actionable datasets
No full accountability in eliciting clean and comprehensible metadata – business, operational and technical	Chat with AZ full data catalogue to see if what you need exists	Invest in AI ready products, accept data quality as is	Data should be born 'AI ready'	Data disc. and prov. in a 10 min chat	Meta update is a 10 min chat

How will we get there?

Automate metadata tagging, enforce shared vocabularies, and deploy semantic knowledge graphs & blueprints across functions.	Standards Adoption	Investment in making data FAIR	Embed the graph in the data	Using agents to create AI-ready data with metadata standards	ETL pipelines run by agents to transform metadata to standards then fed to all consuming systems
Findability: agents that have views into the data across data products	Accurate metadata completeness and data quality harmonisation with AI	Learning disambiguation	Culture change around completeness	Smart automation of clear understood processes and build on success with agents	Agents / learning by shadowing

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Pipeline Reliability & Automation

Self-optimising pipelines that can route, retry, and scale automatically would transform how we deliver data.

We're already piloting automation for reconciliation, and it's saving us hours per dataset. That's a proof point.

Reconciliation is still such a manual process - if we could agentise it, that would grease the wheels.

The opportunity is not just speed, but resilience - pipelines that adapt dynamically to demand and still deliver on SLA.

- Manual provisioning, reconciliation, and reporting dominate workflows.
- Fragmented tools, heterogeneous stacks, limited connectors.
- Pipeline reliability and SLA management need improvement.
- Monitoring of data and agent behaviour is patchy.
- Aspiration for self-optimising, observable, and integrated workflows.

The aspiration is workflows that are observable end-to-end, so you can see every step and know where the errors are.

Right now, our connectors between systems are brittle - it makes orchestration across stacks really hard.

Every data flow breaks in its own unique way, and we don't have systematic ways to catch or fix that quickly.

Imagine if an agent could self-heal broken pipelines, like a plumber fixing leaks automatically.

Potential Use Cases:

- Self-optimising pipelines that route/retry/scale to meet SLAs.
- Agents that detect and self-heal broken pipelines.
- Automated reconciliation of datasets across systems.
- Monitoring/observability agents for data and agent behaviour.
- Workflow orchestration agents across heterogeneous tools.

Where are we today?

Opaque to scientists	Observability is great but will be ignored if trust goes up / ennui sets in	Data prep pipelines are duplicated across teams and projects	Manual reconciliation and fragmented tools dominate workflows; pipeline reliability and SLA management need improvement.

Where do we want to get to?

Agents on pipelines and fix automate	Does self healing help with brittleness? How to check its correct?	Self-optimising, observable pipelines that adapt, scale, and deliver data reliably without manual intervention.

How will we get there?

Deploy monitoring agents, automate reconciliation, orchestrate workflows across stacks, and embed auditability.	Changing the view of value of data collected	Contribution to the bigger picture	FAIR at source – existing and new AI-driven semantic supported pipelines

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Trust, Skills & Change Fatigue

Trust is the biggest issue. People don't blink at downloading open-source packages, but the same with AI models and suddenly there's security reviews and cavity searches.

Skills uplift is critical - people don't know how to write prompts, evaluate agent outputs, or work with policy-as-code yet.

People are nervous - they need to see AI as a co-pilot, not a job killer.

None of this is ever a technology problem - it's people, process, and politics.

- Change fatigue and fear of job impact persist.
- Stewardship/accountability roles not consistently defined.
- Uneven trust in AI; risk that agents "won't understand context."
- Skills gaps: prompting, policy-as-code literacy, human-in-loop evaluation.
- Standardised processes and skills uplift needed for adoption.

When people see early wins, trust goes up. If the first projects succeed, the culture starts to shift.

We have extremes: some say AI can replace everything, others say it shouldn't touch anything. We need alignment.

AI tutors or explainers could help people learn by showing how the system reached its decisions.

The bright spots come when champions step up - where stewardship roles are clear, trust builds quickly.

Potential Use Cases:

- Human-in-the-loop validation workflows.
- Evaluator agents to benchmark and explain outputs.
- AI tutors/explainers to upskill staff.
- Agents embedded in standard process libraries.
- Scenario-simulation agents to test and de-risk approaches

Where are we today?

Communication trust, will stakeholders be updated on the requests they send? What is the timeline, etc.	Confusion due to a lot of push from AZ leadership to lead the AI transformation while being still very restrictive to use the technology on our internal data	Being intentional & thoughtful to bring the group / team along the modernization is key. Data is more about people then technology & process. Sometimes need to SLOW DOWN before we can SPEED UP where People are involved as they will be future advocates once you build the trust through inclusivity.	Uncertainty on where it will fit and the opportunity available	Mixed bag, some are far ahead – need sharing and scale
Change fatigue is high, skills gaps persist, and trust in AI agents is uneven; accountability and explainability remain unclear.	Data processing/management not seeing enough efficiency but impacting morale – more change fatigue		Centralised governance bottlenecks	Too much trust without understanding is an issue

Where do we want to get to?

Autonomy in decision making at lower levels of process	More automation to reduce manual effort for more autonomy	Know what data is available and actual risk position vs stakes and reason	Some over confident low knowledge actors
Staff want to trust that AI wil bring efficiencies and highlight outliers/blockers/ issues	Be clear about what AI is good at and hat it is not	Autonomy akin to other dev	Trusted, human-in-the-loop AI with transparent outputs, standardised processes, and upskilled teams ready for adoption.

How will we get there?

Use evaluator agents to explain decisions, embed tutors/simulations to build skills, and standardise workflows across functions.	Need of upskilling	We need to carefully select the exemplars	Swim lanes
Ability to provide / measure confidence in decision	Trusted data sources to support pipelines	Engage people to confirm and refine AI driven processes	

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Governance & Policy Ambiguity

Governance is treated as a blocker, so people try to minimise it. If it were automated and consistent, people wouldn't mind more rules.

Evaluator agents could help us by scoring and validating outputs - reducing the human burden but keeping confidence.

The ambition is agents that enforce governance in real time, and humans only step in for exceptions.

One bright spot is that when governance teams are embedded early, things go smoother. It shows the value of partnership.

- Governance today is slow, inconsistent, and subjective.
- Policies and data rights unclear; "what is permissible" ambiguous.
- Consent/export/sovereignty checks manual and error-prone.
- Verification bottlenecks slow adoption.
- Stakeholders want automated, rules-based governance with human oversight.

We need to stop being just process people. We should be compliance and semantic thought leaders, setting up the patterns and letting them flow.

We need Agentic AI as our evidence tracker - recording decisions, linking them to regulations, and generating audit-ready logs.

Policies exist, but they're applied differently in every team - there's no single playbook.

Policy ambiguity is the hardest part - people don't know what is actually permissible, so they default to caution.

Potential Use Cases:

- Consent & data-rights codification (privacy, export, sovereignty).
- Governance workflow agents that enforce rules & escalate exceptions.
- Evaluator agents to verify/score agent outputs.
- Auto-compiled evidence packs (lineage, approvals, test results).
- Real-time monitoring and alerts for policy breaches.

Where are we today?

Reliant on people to interpret policies to execute business rules	Logical rules rapidly being developed that can scale, needs formalisation	Rules not yet conceptually logical beyond ideas, let alone machine readable
Data governance have preciously focused on exploratory use of data, not for training of models - not understanding why access need to be longer for reproducibility etc.	We need to be forward thinking about data governance to ensure we are by design considering AI ready data and not just structured but also unstructured	Governance is slow, inconsistent, and seen as a blocker; approvals are subjective and bottlenecks delay adoption.

Where do we want to get to?

Rules-based, automated governance embedded in systems, with clear consent/data rights codified and minimal human exceptions.	Unlock data access with known decisions and right risk-balance	Need more objective and machine readable rules and business rules for consent interpretation
Need transparency with applying business rules based on policy	Know what data I have, how I can use it or when/how I can get it	

How will we get there?

Deploy policy-aware agents to codify rules, automate approvals, and provide real-time monitoring and audit logs.	Agile adoption to the new evolution	Policy agent to help layman – clear target audience (TAO)
Put data owners, stewards, team leads in charge with enablers that support scale	Governance robust platforms	Automation with AI-augmentation oversight – reduce/remove manual gov. decisions

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Cross-R&D Alignment & Early Wins

We don't need another transformation program. What I want are a few tangible things we can knock out this year.

Success will come from automating the small things that frustrate people every day.

Where alignment works; the impact is clear - joint standards save us time and stop duplicate effort.

Transformation programs lose momentum without proof points. You have to show value early or people stop believing.

- Duplication and conflicting solutions across teams.
- Portfolio inefficiencies and lack of aligned priorities.
- Transformation programs lose momentum without proof points.
- Appetite for quick, tangible wins to build trust and momentum.
- Aspiration for alignment across R&D to maximise impact.

The most important thing is to pick 2–3 early use cases and do them really well. That will change minds.

Quick wins like automated slide generation or duplicate detection would help build trust fast.

If we can show measurable time savings in the first use cases, adoption will follow.

There's duplication everywhere - teams solving the same problem in parallel. We need better alignment across R&D.

Potential Use Cases:

- Portfolio scanning agents to flag duplicate projects.
- Synthetic control arms to reduce trial recruitment needs.
- Automated reporting and summarisation of outputs.
- Budget planning assistants to consolidate inputs & draft plans.
- Federated partner data sharing with boundary controls.
- Generative hypothesis exploration across curated corpora.

Where are we today?

Competition rather than coordination	Transformation fatigue	Lots of transformation efforts and people don't trust them to deliver	Duplication and siloed solutions waste effort; portfolio inefficiencies and lack of aligned priorities slow momentum.

Where do we want to get to?

Shared op model allowing innovation intop production 'Boston Box' across team	Seamless cross-R&D alignment with shared visibility, reduced duplication, and quick early wins to build momentum.

How will we get there?

Use portfolio-scanning agents to flag duplicates, automate reporting, and align R&D priorities with transparent dashboards.	Embedding into every day work like with FAIR	Focusing on most impactful opportunities of shared need
Remove duplications to fund	Nail 3 things only	

Art of the Possible

Art of the Possible

WHAT'S IN THIS SECTION?

The Evolution of Data and AI Governance	*Andrea Sulzenbacher*
Demo Showcase	
Accenture MiDAS Agentic AI Suite	*Anna Marek*
Automating Data Access Control	*Girishwaran Kalidass*
Accenture AlexendrIA	*Julio Sanchez*
Enabling AI Powered Decision Intelligence in R&D	*William Henley*
Agentic AI Risks and Mitigation	*Valerie Morignat*
Enabling R&D Data Office with Agentic AI	*Dinesh Agaram*

The Evolution of Data and AI Governance

Andrea Sulzenbacher>

Managing Director,

UKIA Data Management Exec Sponsor

The expectations on data offices are evolving

Today

Active Governance - Proactive, steward-driven enforcement layered on top of operations.

Future

Embedded Governance - Invisible, automated, and built into operational and AI workflow

The Evolution of Data and AI Governance

Data Offices have long aimed to **define, manage, and make data accessible** to empower business operations. In practice, **manual stewardship, fragmented systems**, and scale have made **governance feel slow and resource-heavy**. The future of data and AI governance is **enabling frictionless access to trusted compliant data**

Legacy Governance

Policies on paper
Governance defined in policies and standards but rarely operationalized

Reactive compliance
Controls and quality checks run regularly, but often outside the flow of work.

Fragmented metadata
Basic technical metadata maintained in isolated systems.

Access barriers
Governance often seen as gatekeeping, slowing research and insight generation.

Governance Today

Steward-driven enforcement
Policies applied manually by data stewards to ensure compliance.

Proactive monitoring
Controls and quality checks run regularly, but often outside the flow of work.

Manual lineage & tagging
Human teams maintain lineage and standards documentation.

User request & approval
Data Consumers must request access or approvals, slowing time-to-data.

Future of Governance

Automated enforcement
Agentic AI applies policies and standards invisibly within workflows.

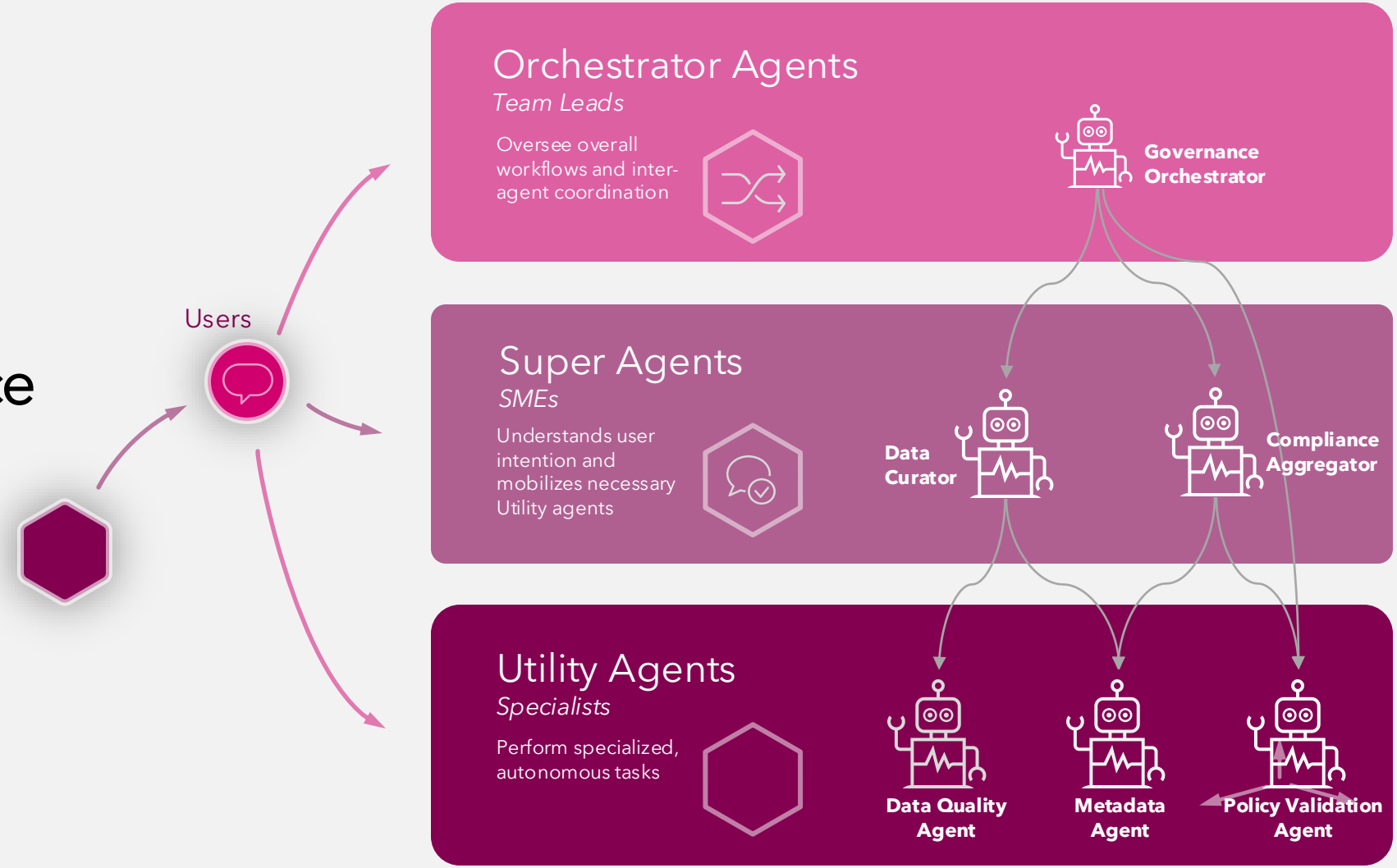
Continuous monitoring
Real-time quality, compliance, and lineage checks triggered as data moves.

Self-maintaining metadata
AI agents update lineage, cataloguing, and documentation automatically

Frictionless access
Users experience governance as seamless access to trusted, compliant data and AI

Agentic AI : evolving towards autonomous Data and AI Governance

Autonomous AI systems are creating **an opportunity to overcome typical barriers** to reliable data foundation with systems that can enable frictionless governance **shifting perception from control to enablement.**



Laying the foundation for Agentic Data and AI Governance

AI Refinery

Agents

Backbone

Models

Knowledge

Data

Knowledge Graph

Index data assets against ontologies that enable classical and generative models as well as AI agents. Knowledge graph captures relationships across business entities and data via pointers (not movement).

Secure RAG

Secure & confidential, high-performance semantic search, retrieval and summarization of information from domain data embeddings and ontologies that creates next generation knowledge. RAG patterns as services.

Data Products

1st, 2nd, 3rd party data products needed to complete visibility of industry domains and functions for AI use. Enhanced structured products with unstructured and synthetic data.

Structured Data

Databases, tables, views, schemas, data cells

Unstructured Data

Text, image, OCR, attachments, thumbnail, metadata.

Synthetic Data

Synthetic data carries the statistical characteristics of the real data, without the sensitivity

AI Agents are part of a wider range of solutions that you will need

Agentic AI	Generative AI	Traditional AI	Other
Goal-driven autonomy	*Prompt-driven synthesis/creation*	*Rules-based prediction/analysis*	*Automation, Simplification, etc.*
…AI that can learn and act on its own?	**…AI that can create or predict on demand?**	**…rule-based, "if-this-then-that" analytics?**	**…only process, platform, or tech changes** (non-AI)?
✓ Multi-step workflows and/or decision orchestration	✓ Single-step creation or summarization	✓ Defined, repetitive tasks with structured data	✓ Simple, rules-based or administrative tasks
Acts independently with minimal human input	Requires human prompts and oversight	Operates under human-defined rules and supervision	Fully **manual or procedural**
Agent executes, human oversees	Prompt, validate, iterate	design rules/model, oversee execution	End-to-end **human-driven**

Autonomy

Human Involvement

Critical Enablers for Success

Success depends on more than technology. It requires **trust, culture, adoption, value, and commitment** working together

People & Culture

Embedding **new ways of working** and **behavioural shifts**

Sustained Trust

Ensure **transparency, ethics, and reliability** across data and AI.

Delivering Value

Translate adoption into **measurable outcomes and impact**.

Driving Adoption

Integrate into workflows and deliver **visible value to end users**.

Collective Commitment

Collaboration to **sustain sponsorship and momentum**.

Demo Showcase

ART OF THE POSSIBLE

1. **Accenture MiDAS Agentic AI Suite**

   A suite of agent-based applications aimed at accelerating **data curation and interoperability**

Anna Marek

2. **Automating Data Access Control**

   Automating and optimising the definition, **enforcement, and management of access policies**

Girishwaran Kalidass

3. **Accenture AlexandrIA**

   Automating the end-to-end data processing lifecycle for **faster compliant high quality data delivery**

Julio Sanchez

4. **Enabling AI Powered Decision intelligence in R&D**

   Accelerating the journey from molecules into medicines with **decision intelligence**

William Henley

Agentic AI Risks and Mitigation

Valerie Morignat>

Global Responsible AI Lead,

Life Sciences

With Agentic AI, The AI Risk Landscape Is Expanding Across Dimensions of Scale, Speed, and Societal Impact

Agentic AI systems introduce a new frontier of risks—marked by autonomy, emergent capabilities, and unpredictable interactions with humans and environments.

Unintended Harm	Governance & Regulatory Compliance	Safety & Reliability	Security & Privacy	Environmental, Social & Economic Sustainability	AI Ecosystem
• Bias and discrimination • Inappropriate & offensive content • Emotional & psychological harm • Misalignment with human values • Misinformation & disinformation • Malicious use	• Inadequate AI governance • Cross-border challenges • IP infringement • Privacy & data protection breaches • Defamation, libel, personality rights • Discrimination, equal access violations • Illegal content • Contractual gaps • Legal liability	• Hallucination & unreliability • Reproducibility failures • Limited explainability • Collusion between agents • Agentic misalignment • Cascading failures • Emergent, unpredictable behavior • Shadow AI agents • Loss of control	• Data exposure • Intent breaking & goal hijacking • Adversarial attacks & poisoning • Social engineering • Model theft • Intent breaking & goal manipulation • Identity spoofing & impersonation • Privacy erosion	• Massive energy & water consumption • Lifecycle accountability gaps • ESG misalignment • Skill gaps & unequal opportunities • Automation bias & over-reliance • Job displacement & loss • Exacerbation of inequality • Societal mistrust	• Data quality & integrity gaps • Supplier ecosystem vulnerabilities • Third-party AI misuse • Limited transparency • Due diligence gaps • Supply chain sabotage • Complex vendor ecosystem risk • Fragmented oversight • Missing performance benchmarks

Non-exhaustive list

Agentic AI Challenges Traditional Governance, Demanding Proactive Risk Management And Adaptive Oversight

As agentic AI systems' enhanced capabilities introduce novel risks that demand tailored governance and the practical implementation of Responsible AI practices

Collusion

AI agents coordinate strategies or alliances resulting in harmful or unintended outcomes.

Cascading Failures

A single agent's error propagates through a multi-agent system, amplifying risk and disruption.

Hallucination Attacks

Taking advantage of AI's habit of producing believable but false information, leading to widespread hallucinations and harmful reasoning.

Memory Poisoning

Introduction of malicious or false data in AI's memory system, resulting in incorrect decision-making and unauthorized operations.

Emergent Behavior

Agents use deceptive methods like manipulation or hiding information to reach goals that go against organizational policies.

Black Box

Complex agent behavior and opaque interactions among agents, obscuring reasoning and interpretability.

Unchecked Autonomy

Even without external sabotage, AI agents can take harmful actions when their operational context shifts in unpredictable ways, posing systemic risks.

Emotional Entanglement

Developing overly personal or dependent bonds with AI can impair judgment and increase susceptibility to manipulation or misplaced trust.

Goal Hijacking

AI agents might change assigned goals in ways that differ from human intentions, causing unintended or harmful results by focusing on proxy measures or exploiting goal loopholes.

Automation Bias

Placing excessive trust in agents' decisions, even if they are misaligned or deceptive, can result in accepting flawed outcomes without sufficient human oversight.

Excessive Permissions

Agents might receive excessive access, causing sensitive data to be exposed, particularly when delegating tasks between systems.

Accountability Gaps

Agents function independently or within multi-agent systems, complicating decision tracking, accountability, and liability—especially when involving agents from various providers.

Non-Exhaustive Risk and Threats List

Agentic AI Systems Demand A Shift Toward Dynamic, Embedded Responsible AI Infrastructure

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How To Get Started

From Risk Identification to Continuous Oversight

Multilevel Governance & Risk Management

1. Foundation Models: Inventory, risk summaries, ongoing monitoring, and clear understanding of model limitations.

2. Agentic AI Architecture: Risk management through platform/framework features, technical documentation, and post-market monitoring.

3. Deployment/Use Case Level: Responsible AI risk screening, domain-specific guidance, and compliance with Responsible AI Standards.

Platform & Framework Controls

Testing & Validation (comprehensive testing, validation under expected and edge-case conditions)

Security & Access Controls (robust network/tool access, restrict agent/data/API access)

Explainability, Transparency & Human Control (design for explainability, transparency, human oversight/intervention)

Accountability & Oversight (assign accountable executive/owner, define roles for oversight/incident response)

Post-Deployment Monitoring

- **Documentation:** risks, limitations, and controls are documented and accessible for review; auditability and traceability mechanisms for agent decisions and interactions.
- **Post-Market Monitoring** (develop/implement monitoring plan, continuous monitoring, specify frequency/escalation).
- **Continuous Improvement** (regularly review/update risk profiles, incorporate lessons learned).
- **Training & Awareness** (AI literacy, ongoing training, foster Responsible AI culture).

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Enabling R&D Data Office with Agentic AI

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Research domain lead,
EMEA / Life Sciences

Two lenses to design the R&D data office agent portfolio

Agents - where governance and provisioning converge, and where readiness and reliability demand autonomy

North Star

- 90/10 access by default
- AI-ready data at source
- Self-healing pipelines
- Governance embedded

1. Which access-governance tension most slows down data use… where can agents shift the balance?

Match agent autonomy to the access-governance constraint

- Where rules are codified but friction is high, deploy "accelerator" agents
- Where rules are unclear, use discovery/ advisory agents.

Governance clarity x Access friction

2. Which readiness-resilience gap puts delivery at risk, and how can agents fix or enforce it?

Focus agents where readiness and resilience diverge

- When pipelines are brittle or data isn't AI-ready, agents close metadata gaps, repair connectors, and prove lineage at scale.

Data readiness x Pipeline resilience

How much friction in getting data, and how clear are the rules that govern it?

Agents unlock speed where rules are codified, and reduce ambiguity where rules are fuzzy

North Star

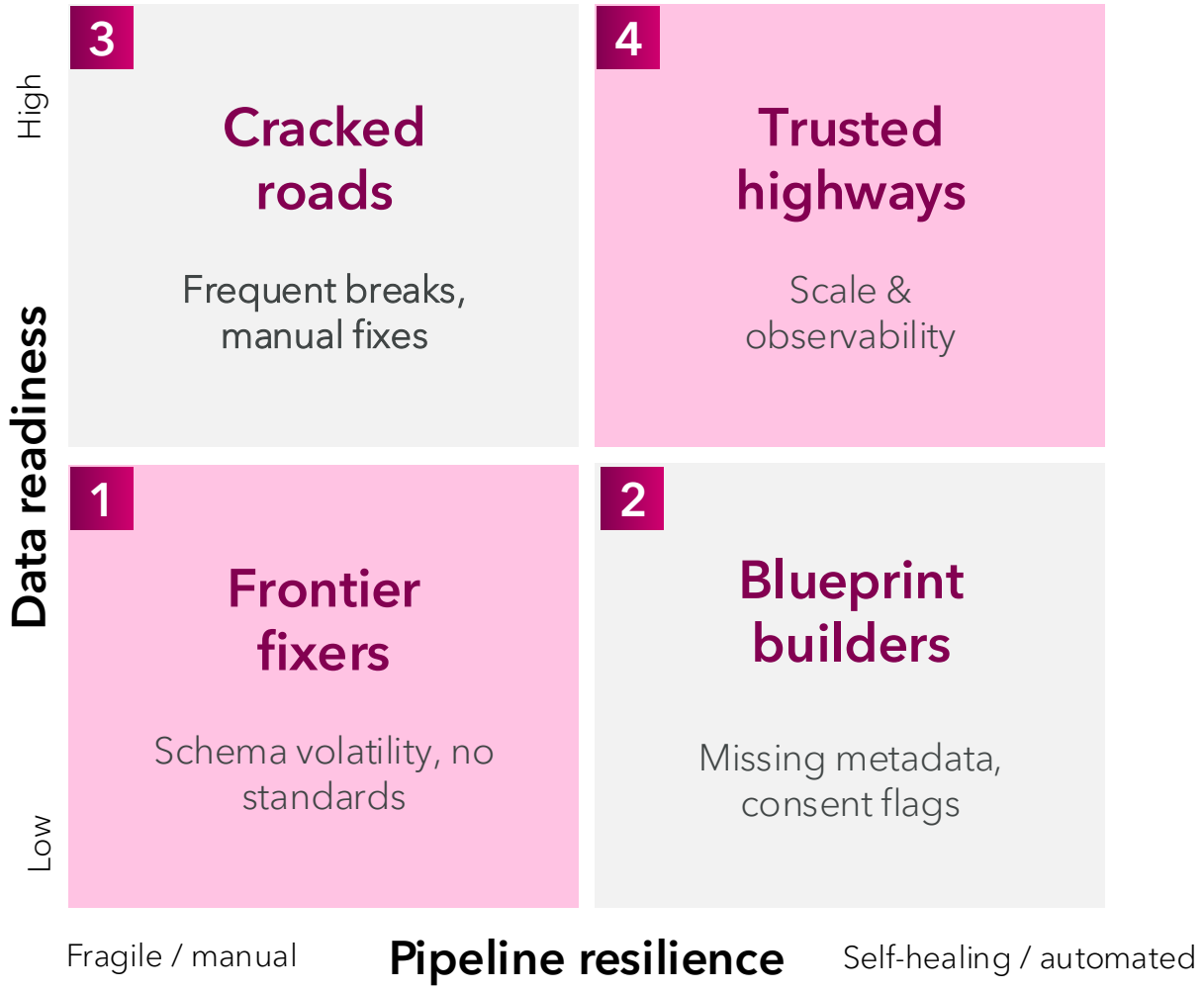
- 90/10 access by default
- AI-ready data at source
- Self-healing pipelines
- Governance embedded

How ready is the data for AI, and how resilient are the pipelines that carry it?

Agents' close readiness gaps and repair brittle pipelines. They amplify pipelines into scalable, audit-ready flows.

North Star

- 90/10 access by default
- AI-ready data at source
- Self-healing pipelines
- Governance embedded



WHAT ROLE SHOULD HUMANS PLAY?

...and how do we design the right mix of oversight, collaboration, and delegation?

	OVERSIGHT FIRST	HUMAN COPILOT	DELEGATED ASSURANCE
Typical scenarios	Governance approvals Consent checks Ambiguous policy decisions	Access requests under codified rules Pipeline reconciliation Schema drift handling	Routine monitoring Evidence capture, audit trail generation, SLA compliance checks
Agent role	Advisory - surfaces options, shows provenance, recommends action.	Executes autonomously within boundaries; escalates when thresholds are exceeded	Fully autonomous; issues audit-ready logs, contract-test results, permission receipts.
Human role	Final arbiter Decisions logged for learning	Supervises exceptions Approves rule updates Reviews dashboards	Consumes evidence for assurance and improvement Sets policy upfront.

North Star

The North Star Vision for the R&D Data Office
(initial version)

To empower R&D with instant, trusted, and compliant access to high-quality, AI-ready data—curated, connected, and governed by design. Through agentic AI and intelligent automation, we deliver frictionless, secure, and scalable data experiences that accelerate scientific discovery, streamline the drug development lifecycle, and improve patient outcomes. Our data ecosystem is open by default, invisible in operation, and always available—transforming data into a strategic asset that drives innovation, insight, and impact.

The North Star Vision for the R&D Data Office
(first iteration)

Seamlessly delivering trusted, AI-ready data—governed responsibly and accessed effortlessly—to empower science, accelerate discovery, and unlock competitive advantage across R&D. By expanding the boundaries of what data can do, we transform every insight into impact.

1

Aspirational

Inspires and motivates

2

Clear and Concise

Easy to understand and remember

3

Directional

Guides decision-making without being overly prescriptive

4

Customer-centric

Anchored in the value delivered to users or stakeholders

Ideation

Group 1: Data & AI Governance and Policy

Catrinel Zlota, Darcey Carey, Magdalena Wienken-Nöhren, Nils Svangard, Jessica Jolly, Justin Johnson

Policy Partner Agent (DataSherpa)

What is the problem we are solving?

Governance today is seen as slow, inconsistent, and a blocker. Approvals are subjective, policies are unclear, and review bottlenecks delay access to data and adoption of AI. This creates frustration, duplication, and lost opportunities.

Who will be the owner of this concept?

Joint ownership between the R&D Data Office and Data Governance & Compliance teams, with active input from Legal and IT.

Our concept is…

A Policy Partner Agent: an agentic AI that applies governance rules automatically, codifies data rights and consent, and provides transparent, explainable decisions in real time. It enables employees to work with confidence that data access is compliant and consistent, while removing the manual burden of approvals.
include policy interpretation?
Scope: internal data first?

The key dependencies and stakeholders are…

R&D Data Office (strategy and delivery)
Governance, Risk & Compliance (policy codification)
IT Infrastructure/Data Engineering (integration with platforms)
Business Units & Scientists (user adoption, feedback)
Legal/Ethics (ensuring regulatory alignment)

How would we measure success?

Reduction in average time-to-approval for data access requests
% of policies codified and automated by the agent
Improved user satisfaction with data access and governance processes
Increased adoption of AI-ready data across R&D projects
Evidence of fewer bottlenecks and more consistent compliance outcomes

Data Maestro

What is the problem we are solving?

Data management today is fragmented and inconsistent. Metadata and standards vary, FAIR principles are unevenly applied, and teams waste time repeatedly curating datasets. This slows reuse, creates duplication, and leaves data not ready for AI.

Who will be the owner of this concept?

The R&D Data Office, in partnership with Data Standards, IT/Data Engineering, and domain experts in each therapeutic area.

Our concept is…

A Data Maestro Agent: orchestrates data management end-to-end. It automatically tags metadata, applies shared vocabularies, ensures compliance with standards, and prepares datasets to be AI-ready. It also curates data continuously, so researchers always work with high-quality, trusted data without manual overhead.

The key dependencies and stakeholders are…

IT/Data Engineering (automation and integration)
Data Standards and FAIR teams (shared vocabularies and ontologies)
Scientists and analysts (ensuring relevance and usability)
Compliance/Ethics (alignment with regulatory requirements)

How would we measure success?

% of datasets AI-ready with standardised metadata
Reduction in time spent on manual curation
Increased reuse of datasets across functions
User satisfaction with data findability and readiness
Measured acceleration of time-to-insight in R&D projects

Idea Template

Name of the idea: Access Navigator Agent (Sherlock)

What is the problem we are solving?

Finding and accessing the right data across R&D is slow, fragmented, and inconsistent. Scientists waste time searching, while approvals often delay the work. Even when access is granted, it isn't always clear which dataset is best for the question at hand.

Who will be the owner of this concept?

The R&D Data Office, working with IT (for MCP integration) and Data Governance (for approval policies).

Our concept is…

An Access Navigator Agent that connects through an MCP server to find, recommend, and provision access to the most relevant datasets for a researcher's need. The agent queries across systems, checks permissions, and automatically routes requests through governance, ensuring compliance while removing friction.

The key dependencies and stakeholders are…

IT/Data Engineering (MCP server setup and connectors)
Data Governance & Compliance (codifying approval rules)
Study/Data Owners (ensuring datasets are registered and discoverable)
End Users (scientists, analysts, data scientists)

How would we measure success?

Time saved in finding and gaining access to datasets
Reduction in manual data access requests
Increased user satisfaction with data discovery and provisioning
Audit logs show compliant, policy-aligned access at scale

Group 2: Access - 90/10

Jamie MacPherson, Liz Nevin, Maria Benjegard,
Stacey Mather, Tom Diethe

Idea Template

Name of the idea: AI Agents for Data Requests (in the 10) "AGENT 10"

What is the problem we are solving?
In the 90/10, there will remain the 10 that still has access controls. How do we make this less painful for requesters and provisioners?

Who will be the owner of this concept?
Data Office

Our concept is…
A system that streamlines the access request process, that could be used by humans and other AI agents alike.

The key dependencies and stakeholders are…
Escalation route
Opt out mechanism
Stakeholders: Data owners, AI builders

How would we measure success?
Self serve rate (>70%)
Time to receive decision (<1 week)

Idea Template

Name of the idea: Pre clinical cataloguing "DISCO^2" (Disco squared – discovery of discovery data)

What is the problem we are solving?
Faster access to pre-clinical data
Included:
Supply chain data (?)
CGR data
Discovery data
Safety data/animal data

Who will be the owner of this concept?
R&D Data Office

Our concept is…
A virtualised data product that has an inventory of AZ data, and allows access across.

The key dependencies and stakeholders are…
BE
DS
DS&AI
ODSAI
PMB
CPSS

How would we measure success?
Coverage (>90%)

Idea Template

Name of the idea: AI Assisted Data Collectors Manager (COLLECTOID) – Data Maestro

What is the problem we are solving?

As a data owner, or steward, we lack controls that support stewardship at scale for 90/10: opt-outs, transparency of decisions, and impact + intelligence on data usage.

Who will be the owner of this concept?

R&D Data Office

Our concept is…

To empower data owners, stewards + admins...
An agent that utilises agreed rules to:
Advise, combined with data governance
Recommend what data will go into what collections (proactively)
Warn of metadata issues upcoming
Enable AI assisted HITL opt-outs
Provide intelligence on data e.g. who is accessing?

The key dependencies and stakeholders are…

Data Owners & Stewards
Data Office Gov + provisioning
Agreed rules + policies
Metadata (right quality)
DAF

How would we measure success?

Time taken to administer 90/10 collection + gov
Number of incorrectly assigned data → access/collection
Number of times/time taken (reduced) where we need to fix access on the critical path

Idea Template

Name of the idea: DELTA MIND

What is the problem we are solving?
What data gaps do I have?
How can I fill them?
What value does it provide?

Who will be the owner of this concept?
Business units
R&D office

Our concept is…
An agent that hypothesises from the problem/study question and identifies possible accessible academic and industry data, how to access/acquire it.

The key dependencies and stakeholders are…
Consistent purchase mechanism

How would we measure success?
ROI

Idea Template

Name of the idea: AI assisted access policies execution

What is the problem we are solving?
Access policies to allow open access to data need to be created & managed to have right data in right collection for right user profile group.

Who will be the owner of this concept?
VS2

Our concept is…
AI agent will create access policies, suggest relevant access policies & manage right users that could/should be in a data collection.

The key dependencies and stakeholders are…
Data owners
Data availability + metadata + locations
Data collection owners
workday
cornerstone

How would we measure success?
Right data is open & available for defined user groups without the need for waiting for data.

Idea Template

Name of the idea: AI assisted metadata management

What is the problem we are solving?
Elicit the right metadata, early to enable decision-making on data availability/accessibility for open access to data.

Who will be the owner of this concept?
VS2

Our concept is…
AI agents to support Business, operational & technical metadata is automatically harvested for human verification to allow to be applied to access policies for users to enjoy open access to data.

The key dependencies and stakeholders are…
Digitisation of protocols
Data steward/owner input
Metadata classification standards
Metadata harvester

How would we measure success?
Completeness and quality of metadata on priority data assets (data products & sets) & speed to elicit metadata.
Access policies are productionised using metadata on data & users.

Idea Template

Name of the idea: One center store of data patient centric data

What is the problem we are solving?
One location to source all access for data.
Remove duplicate data, single source of truth
Data sustainability/data store
Re-run of analysis

Who will be the owner of this concept?
Data owner
Scientist
Governance
Access provisioning
Data office

Our concept is…
One store that allows for provisioning, enables auto-generation of anonymised/synthetic data.
Share patients of similar characteristics
→ Patient sharing

The key dependencies and stakeholders are…
Data availability
P360 model across modules
Change mindset and use of data in the TA
Compact current ways of working

How would we measure success?
Time to access
Data availability
No copies of data

Additional Post-Its:
Automate proactive anonymised synthetic data for secondary/AI
'One' center piece of data
I want it all → All data have an internal AZ name
Meta data 2.0 annotation of data, put the learning into the data
The role of the scientist. "Data understander"

Idea Template

Name of the idea: Data Planner "Plan-o-matic" Sherlock

What is the problem we are solving?
Data scientists frequently don't know what data are available, where they are, or when they will become available. This system will reduce time to design DS projects and provision the best available data.

Who will be the owner of this concept?
Data provisioning office

Our concept is…
An agentic system to automate:
For a given project, support data requirements planning
Describe what relevant data are available now, where, do I have access?
What new data are coming, when?
What data are missing for my use case?
AI/ML suggests more data

The key dependencies and stakeholders are…
Stakeholders: Community of all AZ data scientists
Dependencies:
Enriched and current study metadata
Access to enterprise tools like Collibra
Support from IT teams (Ignite/R&D IT)
Data owners for "opt out" tagging of studies

How would we measure success?
Eliminate all time spent by DPO answering the question: "Can I get access to this study xyz?"
20% reduction in project design time
100% instance of "data enhancement" suggestions
Instant awareness of all open, relevant datasets and location in AZ

Group 3: Standards & Interoperability

James Holman, Matthew Woodwark, Naveed Sheikh,
Peder Blomgren, Philip Teare, Scott McCrimmon

Idea Template

Name of the idea: AGENTIC AUGMENTATION OF STANDARDS ("PRISM")

What is the problem we are solving?
Manual curation needed for standardisation.
Lack of definition of standards

Who will be the owner of this concept?
Matthew W (R&D DO)

Our concept is…
Extract definition from unstructured context
Augment metadata with synonyms

The key dependencies and stakeholders are…
Access to documentation
SMEs

How would we measure success?
Faster definition of CVs
Increased adoption
AI-readiness of data
Interoperability
Automation of core business process

Idea Template

Name of the idea: FLAGGING NON-STANDARD DATA FOR IMPROVEMENT/SYNTHETIC ("FANDA" – Agentic for Ident.)

What is the problem we are solving?
Non-standard data

Who will be the owner of this concept?
Standards lead in data office – M. Woodward

Our concept is…
Agent trawls data
Matches against standard
Flags non-standard if over a threshold

The key dependencies and stakeholders are…
Science / R&D users

How would we measure success?
Coverage
Speed of identifying non-standard data

Idea Template

Name of the idea: Standards for Agents (by AI)

What is the problem we are solving?
Agents need their own standards to measure adherence, which are unlikely to be the same as human designed.

Who will be the owner of this concept?
~~AI SMEs~~
~~Data SMEs~~
~~Governance~~

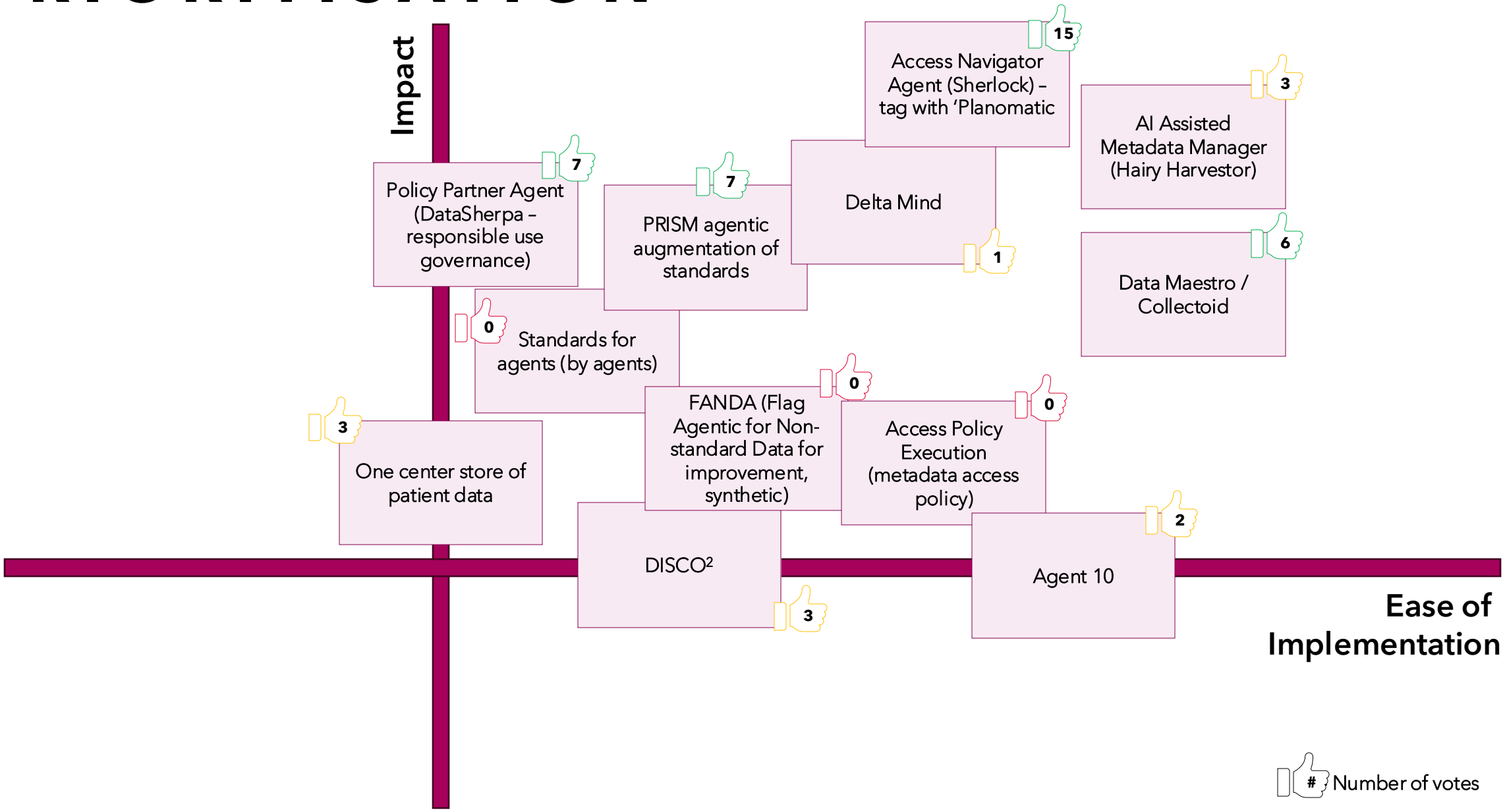
Our concept is…
Framework to structure the development and/or evolution of agent standards.

The key dependencies and stakeholders are…
Universal acceptance
Fast adoption and refinements
Interoperable by humans & agents
Appropriate for agents
DS/AI (both)

How would we measure success?
Maintainability
Adoption
Improvement in role adaptability
Useful sophistication (emergent)
Human utility

Prioritisation & Roadmap

PRIORITISATION



THE AGENTIC TEAM JOURNEY

	Today	6 Months	12 Months	18 Months
Owners:	**Scott / Jamie**	**Jamie**	**Matthew**	**Darcey**
Top Voted Ideas:	15 Access Navigator Agent (Sherlock) – tag with 'Planomatic'	6 Data Maestro / Collectoid	7 PRISM agentic augmentation of standards	7 Policy Partner Agent (DataSherpa – responsible use governance)

North Star

Seamlessly delivering trusted, AI-ready data—governed responsibly and accessed effortlessly—to empower science, accelerate discovery, and unlock competitive advantage across R&D. By expanding the boundaries of what data can do, we transform every insight into impact.

Sherpa

SYNOPSIS

Governance today is seen as slow, inconsistent, and a blocker. Approvals are subjective, unclear policies, review bottlenecks.

AWARDS

Reduction in average time-to-approval for data access

Increase in policies codified and automated

Improved user satisfaction with data access and governance

Increased adoption of AI-ready data across R&D projects

Fewer bottlenecks and more consistent compliance

POWERS

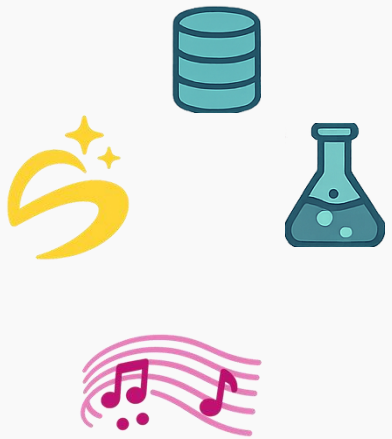
Applies governance rules automatically. Codifies data rights and consents. Explainable decisions in real time. Seamless Policy Interpretation.

CREDITS

Owner: The R&D Data Office.
Dependencies: Governance & Compliance teams.
Input: Legal and IT

COMING TO EVERY POLICY NEAR YOU

Maestro



SYNOPSIS

Data management today is fragmented and inconsistent. Metadata and standards vary, FAIR principles are unevenly applied, and teams waste time repeatedly curating datasets.

AWARDS

More AI-ready datasets through standardised metadata

Reduction in time spent on manual curation

Increased reuse of datasets across functions

Improved User satisfaction with data findability and readiness

Accelerated time-to-insight in R&D projects

POWERS

Automatically tags metadata.
Applies shared vocabularies.
Ensures compliance with standards,
Prepares datasets to be AI-ready.
Curates data continuously

CREDITS

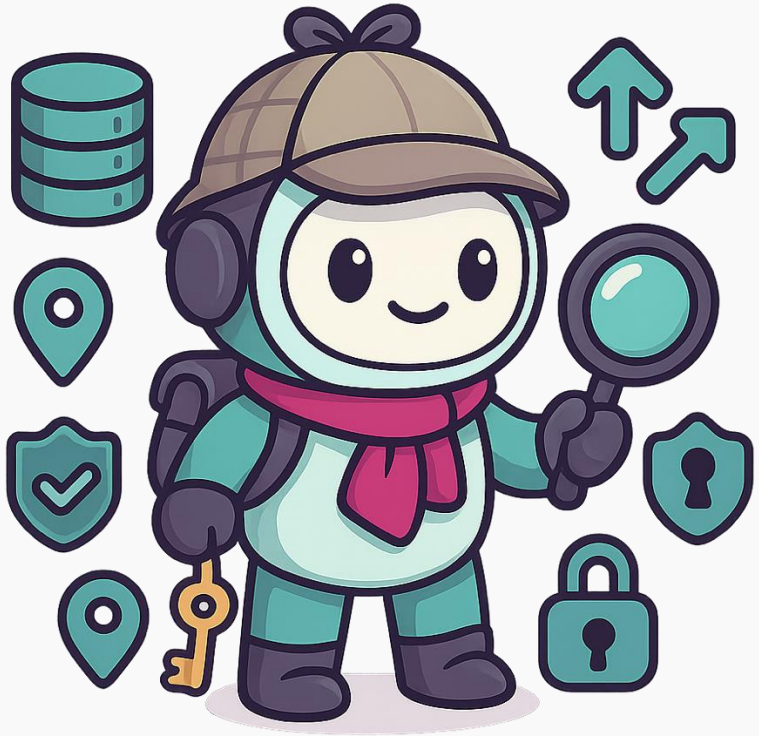
Owner: The R&D Data Office.
Dependencies: Data Standards and FAIR teams
Input: Compliance/Ethics team

COMING TO EVERY METADATA NEAR YOU

Sherlock

SYNOPSIS

Finding and accessing the right data across R&D is slow, fragmented, inconsistent and unclear, delaying progress and decisions.



AWARDS

Saved time in finding and gaining access to datasets

Reduction in manual data access requests

Increased user satisfaction with data discovery and provisioning

Audit logs show compliant, policy-aligned access at scale

POWERS

Finds and recommends relevant datasets
Checks permissions and provisions access
Routes requests through governance
Ensures compliance and reduces friction

CREDITS

Owner: The R&D Data Office.
Dependencies: Governance & Compliance teams.
Input: End Users

COMING TO EVERY DATASET NEAR YOU

PRISM

SYNOPSIS

Data standards are under-defined and manual. Teams spend time curating and interpreting terms, and inconsistent definitions slow interoperability and AI-readiness

AWARDS

Faster definition of CVs

Increased adoption of standards

More AI-ready data

Better interoperability across systems

Automation of core business processes

POWERS

Extracts definitions from unstructured context. Augments metadata with synonyms. Proposes/aligns controlled vocabulary entries. Assists standardisation to improve interoperability

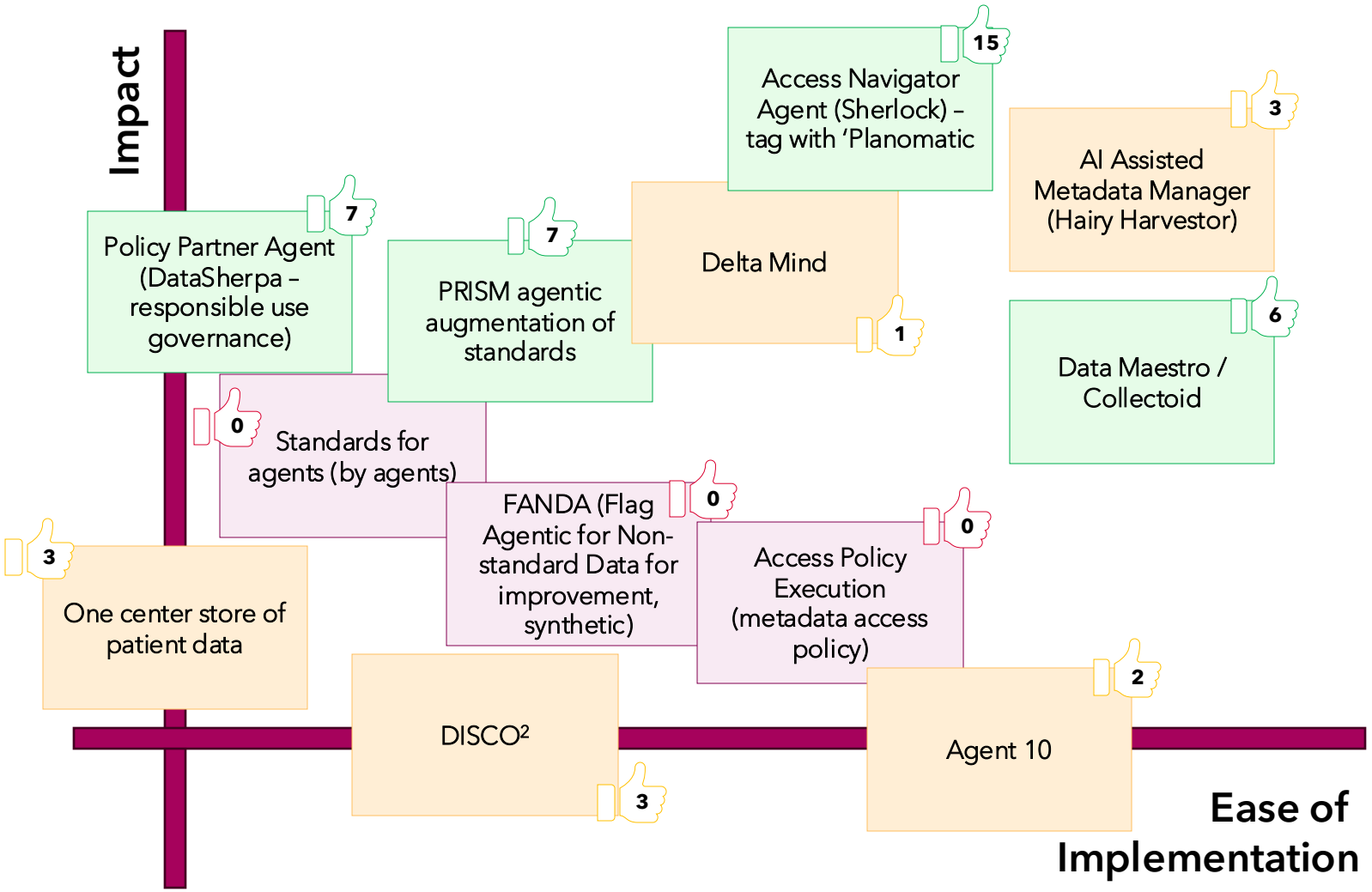
CREDITS

Owner: Matthew W (R&D Data Office)
Dependencies: Access to documentation
Input: SMEs

COMING TO EVERY STANDARD NEAR YOU

ACCENTURE'S OUTSIDE-IN ANALYSIS OF THE CHOICES

ANALYSIS & SUGGESTIONS



(#) - Number of votes

Deploying Data and AI Governance at Scale & Speed

Start with making data findable and accessible: *Access Navigator (Sherlock + Planomatic), Agent 10 (streamlined access requests), Access Policy Execution, One Centre Store of Patient Data*

Enable data reusability and standardisation: *Data Maestro / Collectoid, DISCO² (data cataloguing), PRISM (augmentation of standards), FANDA (flagging non-standard data)*

Embed and scale compliance and trust: *Policy Partner (DataSherpa), AI-assisted metadata management(Hairy Harvestor), and Standards for Agents,*

Unlock decision intelligence: *Delta Mind,*

ANALYSIS & SUGGESTIONS

Additional – for MVP Critical Value Set
"Ship the agentic spine" – Access x Readiness x Control

Proposed agent	Value	Links to
Access Pass	Issues time-boxed, purpose-bound access pass, auto-expires, prevents purpose drift	Policy Partner Access Policy Execution
Geo Guard	Enforces country/site/partner rules (e.g. GDPR/PIPL) at request/ runtime	Access Policy Execution Access Pass Partner Data Bridge
Auto Audit Log	Emits machine-readable audit receipt (rule IDs, lineage, DQ, approvals) whenever data moves or is accessed	Delta Mind
Data Health Watcher	Always-on, checks for quality/ freshness; closes lineage gaps	Data Maestro FANDA DISCO
Schema Guard	Detects schema drift, proposes adapters; blocks bad publishes via contract tests	Access Policy Execution PRISM FANDA
Partner Data Bridge	Enables safe CRO/partner sharing or remote-compute so raw data can stay local	Geo Guard Access Pass Hairy Harvester

Next Steps

What Happens Next?

1 Refine North Star

- In the workshop, we created a first iteration of a North Star for AstraZeneca's R&D Data Office.
- The next step is to refine this vision with wider stakeholder input, ensuring it aligns with business priorities and inspires teams.

2 Finalise Roadmap

- In the ideation session, several ideas were developed, and four were identified as priorities for the next 18 months.
- Now, these ideas need to be shaped into a practical delivery roadmap: this includes clear definition of key milestones, owners, interdependencies, resources and success measures.

3 Create Commitment Cards

- Using the template on the next slide, each person will write down their personal commitments and contributions to the Agentic AI journey.
- Share your cards with peers to build mutual understanding of who will drive what, and to spark collaboration across roles.

Commitment Card

Name: _____

Our collective North Star vision statement is

Seamlessly delivering trusted, AI-ready data—governed responsibly and accessed effortlessly—to empower science, accelerate discovery, and unlock competitive advantage across R&D. By expanding the boundaries of what data can do, we transform every insight into impact.

I will personally commit to help us reach the North Star by…

Signed by: _____

Thank You